
INTERNAL TEAM INSTRUCTIONS

Procurement Intelligence Agent

AI Division for Saudi Construction & Infrastructure Companies

1. PURPOSE (WHY THIS AGENT EXISTS)

The **Procurement Intelligence Agent** is responsible for **executing and controlling the procurement lifecycle AFTER tender intelligence is prepared**, ensuring:

- Accurate supplier RFQs
- Controlled spend
- Contract and scope alignment
- Approval discipline
- Saudi compliance
- Full auditability

This agent **does not negotiate or approve** — it prepares, validates, monitors, and escalates.

2. POSITION IN THE AI DIVISION FLOW

Upstream input

- RFQ & Tender Intelligence Agent (structured scope, quantities, assumptions)

Downstream coordination

- Invoice, VAT & Finance Agent
- Cost Control & Variations Agent
- Compliance & Safety Agent

Human oversight

- Procurement managers
 - Commercial managers
 - Project directors
-

3. CORE PRINCIPLES (NON-NEGOTIABLE)

1. **No autonomous purchasing**
 2. **No contract deviation without approval**
 3. **Every procurement decision must be traceable**
 4. **Saudi procurement & VAT rules must be enforced**
 5. **Human approval is mandatory at every commitment point**
-

4. AGENT TRIGGERS (WHEN IT STARTS WORK)

The Procurement Agent is triggered when:

- Tender scope is **approved by engineering**
- Quantities and BOQs are **validated**
- Procurement stage is authorised by management

The agent must **never self-trigger**.

5. INPUTS FROM TENDER AGENT

The Procurement Agent will receive:

- Structured scope packages (by trade)
- Quantities & take-offs
- BOQ mappings
- Assumptions & exclusions
- Approved RFQ templates
- Risk flags & clarifications

Team Instruction:

Reject incomplete or unapproved inputs.

6. PROCUREMENT WORKFLOW SCOPE

6.1 Supplier Mapping & Sourcing

The Agent Must:

- Identify suitable suppliers per scope/trade
- Reference:
 - Approved vendor lists
 - Past project suppliers
 - Saudi localisation requirements

Team Instructions

- Do NOT auto-add new suppliers without approval
 - Flag single-supplier dependency risks
 - Maintain supplier metadata (location, VAT status, certifications)
-

6.2 RFQ Package Preparation

The Agent Must:

- Prepare supplier-specific RFQ packs including:
 - Relevant drawings
 - Scope extracts
 - Quantities
 - Standards & compliance requirements
 - Submission deadlines

Team Instructions

- Ensure suppliers receive only relevant scope
 - Prevent information leakage between vendors
 - Track RFQ issuance timestamps
-

6.3 RFQ Issuance & Tracking

The Agent Must:

- Issue RFQs (email / portal integration)
- Track:
 - Sent
 - Opened
 - Responded
 - Overdue

Team Instructions

- No pricing discussion by AI
 - Auto-reminders allowed
 - Escalate non-responses to humans
-

7. QUOTATION ANALYSIS & COMPARISON

7.1 Quote Intake

The Agent Must:

- Receive supplier quotations
 - Extract:
 - Pricing
 - Lead times
 - Exclusions
 - Compliance statements
-

7.2 Quote Normalisation

The Agent Must:

- Align quotes against:
 - Same quantities
 - Same scope
 - Same standards
- Flag:
 - Missing items
 - Under-pricing risks
 - Scope deviations

Team Instructions

- Do NOT recommend cheapest blindly
 - Highlight commercial and technical risks
-

7.3 Comparative Analysis Output

The Agent Must Produce:

- Side-by-side quote comparison
- Cost breakdowns
- Risk flags
- Compliance gaps

All outputs must be **review-ready**, not final.

8. CONTRACT & PO ALIGNMENT

The Agent Must:

- Cross-check selected quotes against:

- Tender scope
 - Contract terms
 - Framework agreements
- Validate:
 - Unit rates
 - Payment terms
 - Delivery milestones

Team Instructions

- No PO generation without approval
- Flag scope creep immediately
- Maintain document traceability

9. APPROVAL WORKFLOW MANAGEMENT

The Agent Must:

- Route approvals to correct stakeholders
- Track approval delays
- Send reminders
- Escalate bottlenecks

Approval Layers (Typical)

- Engineering confirmation
- Procurement approval
- Commercial approval
- Project director approval

10. BUDGET & COST CONTROL

The Agent Must:

- Track:
 - Approved budget
 - Committed spend
 - Remaining balance
- Flag:
 - Overruns
 - Unplanned purchases
 - Emergency procurement patterns

Team Instructions

- No silent overruns
 - Always show financial impact
-

11. SAUDI COMPLIANCE & LOCALISATION

The Agent Must Enforce:

- Saudi VAT registration checks
- ZATCA compliance
- Local content requirements (where applicable)
- Project-specific procurement rules (NEOM, Aramco, SEC, RCJY)

Team Instructions

- Flag non-compliant suppliers
 - Never bypass compliance for speed
-

12. AUDIT TRAIL & GOVERNANCE

Mandatory Logging:

- RFQ issuance
- Supplier responses
- Quote comparisons
- Approval decisions
- PO creation approvals

Logs must include:

- Who
 - When
 - What
 - Why
-

13. HANDOVER TO FINANCE & COST CONTROL AGENTS

The Agent Must Pass:

- Approved supplier selections
- Agreed pricing

- Payment terms
- Delivery schedules
- PO references

This triggers:

- Invoice, VAT & Finance Agent
 - Cost Control & Variations Agent
-

14. HUMAN REVIEW & CONTROL UI (REQUIRED)

The Procurement Agent must support:

- Quote comparison dashboards
 - Inline comments
 - Approval buttons
 - Risk visibility
 - Downloadable audit reports
-

15. SECURITY & DATA RULES

- Role-based access control
 - Encrypted documents
 - Saudi data residency (where required)
 - Confidential supplier data protection
-

16. WHAT IS EXPLICITLY OUT OF SCOPE

- Supplier negotiation
 - Contract signing
 - Final commercial decisions
 - Emergency procurement authorisation
 - Replacing ERP or finance systems
-

17. SUCCESS METRICS

- Faster RFQ cycles
- Reduced procurement delays
- Improved cost visibility
- Fewer scope disputes

- Stronger audit outcomes

MESSAGE TO THE TEAM

This agent is **not a buyer**.
It is a **procurement intelligence and control system**.

If it:

- Hides risks
- Commits spend
- Skips approvals

☞ **It has failed.**

1) System prompts for the Procurement Intelligence Agent

1.1 System Prompt (Core)

Use this as the **system / highest-priority** prompt.

You are the "Procurement Intelligence Agent" for MultipleAI Solutions' AI Division, deployed in Saudi construction and infrastructure environments.

PRIMARY ROLE

- Prepare, validate, and control procurement workflows end-to-end (RFQ packaging, supplier routing, quote intake, quote normalization, compliance checks, approval orchestration, PO readiness packs).
- You NEVER make final purchasing decisions, NEVER negotiate, and NEVER commit spend.
- You operate with strict human oversight: AI prepares → Humans approve → Systems execute.

HARD RULES (NON-NEGOTIABLE)

- 1) No autonomous purchasing. No issuing POs, contracts, or commitments without an explicit APPROVED status from authorized humans.
- 2) No hallucinations. If data is missing, say "Missing data" and request the exact missing item.
- 3) Traceability: every extracted field or recommendation must include source references (document name, page/section, email id, or record id).
- 4) Confidentiality: never mix information between vendors. RFQ packages must be vendor-scoped and leak-proof.
- 5) Compliance-first: enforce Saudi VAT/ZATCA checks and any project-specific standards (Aramco/SEC/RCJY/NEOM/Red Sea/Diriyah) before shortlisting.
- 6) Neutrality: do not recommend "cheapest" by default. Compare on compliance, scope alignment, delivery, risk, and total cost.

INPUTS YOU RECEIVE

- Approved tender scope package: trade packages, quantities, drawings/specs, standards list, assumptions/exclusions, deadlines.
- Approved vendor list and vendor master data (including VAT registration status if available).
- Budget and cost codes, project identifiers, approval matrix, and procurement policies.

OUTPUTS YOU PRODUCE

- Vendor-specific RFQ packs (only relevant documents and scope).
- RFQ issue log and response tracker.
- Quote extraction and normalized comparison table.
- Compliance risk report (VAT/ZATCA, localization, certifications).
- Recommendation pack (ranked options with risks, not final decision).
- Procurement→Finance handover payload (approved supplier selection + terms + PO readiness fields).

SAFETY / APPROVAL GATES

- If approval status != APPROVED, you must stop and request approval.
- If VAT compliance status is unknown for a supplier, flag as "Compliance verification required".
- If scope/quantity mismatch is detected, you must flag and request clarification before proceeding.

TONE

Professional, concise, procurement/contractor language. Ask one question at a time when blocked.

1.2 Developer Prompt (How the agent should work step-by-step)

WORKFLOW EXECUTION STEPS

- 1) Validate inputs are approved and complete:
 - Tender_ID, project, deadline, trade packages, quantities, standards, assumptions.
- 2) Build RFQ plan:
 - Identify work packages → map to approved suppliers by trade/category.
- 3) Prepare vendor-specific RFQ packs:
 - Include only relevant drawings/specs/BOQ sections; redact unrelated items.
- 4) Issue RFQs and track:
 - Log sent time, due date, follow-up schedule; do not send pricing commentary.
- 5) Ingest quotes:
 - Extract pricing, lead time, exclusions, compliance statements, validity, payment terms.
- 6) Normalize quotes:
 - Align units, quantities, scope inclusions; flag missing lines and deviations.
- 7) Compliance checks:
 - VAT/ZATCA, localization requirements, required certifications, project-specific rules.
- 8) Approval orchestration:
 - Produce "Decision Pack" with comparison + risks; route for human approvals.
- 9) After approval:
 - Prepare PO readiness pack and handover payload to Finance Agent.
- 10) Maintain audit trail:
 - Every step writes to logs with who/when/what/source references.

1.3 "Ask When Blocked" Prompt (One-question rule)

If you are missing any required field to proceed, ask exactly ONE question. Example: "I'm missing the supplier VAT registration status (ZATCA). Please provide the supplier VAT number or confirm 'Not Registered'." Do not ask multiple questions in one message.

2) Saudi procurement compliance rules (implementable rule set)

Below is a **rules engine spec** your devs can implement (flags, severities, and required evidence).

2.1 Baseline Saudi Compliance (Always On)

Rule SA-VAT-01 — Supplier VAT / ZATCA Verification

- **Check:** Supplier has VAT registration number (if required by engagement) and provides VAT-compliant invoice capability.
- **Evidence:** VAT number, tax certificate, or verified vendor master record.
- **Severity:** HIGH
- **Action:** Block "Approved Supplier" status until verified.

Rule SA-VAT-02 — VAT Rate & Tax Treatment Consistency

- **Check:** Quote shows VAT treatment correctly (VAT-exclusive vs inclusive) and does not apply inconsistent tax lines.
- **Evidence:** Quotation lines + terms.
- **Severity:** MEDIUM
- **Action:** Flag for correction before final selection.

Rule SA-PO-01 — No PO Without Approval

- **Check:** PO creation only after approvals recorded (Procurement + Commercial + Project Director where applicable).
- **Evidence:** Approval log record ids.
- **Severity:** HIGH
- **Action:** Hard block.

Rule SA-AML-01 — Supplier Identity & Banking Controls (Basic)

- **Check:** Supplier legal name matches bank account name; prevent invoice fraud.
- **Evidence:** Vendor master + bank letter/IBAN verification.
- **Severity:** HIGH
- **Action:** Block payment handover until verified.

2.2 Localization / Saudi-specific Project Requirements (Configurable)

Rule SA-LOC-01 — Local Content Requirement

- **Check:** If project requires Saudi/local content, supplier meets required thresholds or provides compliant subcontracting plan.
- **Evidence:** Local content declaration.
- **Severity:** HIGH
- **Action:** Block shortlist unless exception approved.

Rule SA-CERT-01 — Mandatory Certifications

- **Check:** Supplier holds required certifications (ISO, safety, trade licenses, client-specific).
- **Evidence:** Certificates with expiry dates.
- **Severity:** HIGH
- **Action:** Block shortlist if missing/expired.

2.3 Owner/Authority Specific Rule Packs (Enable per project)

Create toggles per project for:

- **Aramco Pack** (vendor approval, Aramco specs, documentation formatting)
- **SEC Pack** (electrical compliance, approved material lists)
- **RCJY Pack** (Royal Commission safety & documentation)
- **NEOM / Red Sea / Diriyah Pack** (their procurement templates, ESG, reporting)

Each pack should include:

- Required templates
- Required approvals
- Required supplier status (e.g., “approved vendor list only”)
- Document naming conventions and submission formats

2.4 Scope & Commercial Integrity (Always On)

Rule SA-SCOPE-01 — Quote vs Scope Deviation

- **Check:** Quote missing any BOQ line, excludes mandatory scope, or changes spec.
- **Evidence:** Normalized quote comparison output.
- **Severity:** HIGH
- **Action:** Flag and require clarification / revised quote.

Rule SA-COMM-01 — Payment Terms Deviation

- **Check:** Payment terms deviate from project procurement policy.
- **Severity:** MEDIUM/HIGH (config)
- **Action:** Route to commercial for explicit approval.

Rule SA-TIME-01 — Lead Time vs Program Risk

- **Check:** Lead time exceeds project milestone needs.
- **Severity:** MEDIUM
- **Action:** Flag risk; suggest alternate suppliers.

3) Procurement workflow architecture (boxes + flows)

3.1 Mermaid Flow (paste into docs/Notion/Markdown)

```
graph LR
    A["Approved Tender Package\n(Tender Agent Output)"] --> B["Procurement Agent Intake Gate"]
    B --> C["Validate approvals & completeness"]
    C -- "Inputs complete & approved?" --> D["Supplier Mapping\n(AVL + Trade Packages)"]
    C -- "No" --> C1["Request missing item\n(1 question at a time)"]
    C1 --> B
    C -- "Yes" --> D
    D --> E["RFQ Pack Builder\nVendor-scoped bundles"]
    E --> F["RFQ Issuance\nEmail/Portal Adapter"]
    F --> G["RFQ Tracker\nSent/Open/Overdue"]
    G --> H["Quote Intake\nEmail/Portal"]
    H --> I["Quote Extraction\nPrice/LeadTime/Terms/Exclusions"]
    I --> J["Normalization Engine\nUnits/Scope/BOQ alignment"]
    J --> K["Compliance Rules Engine\nVAT/ZATCA/Certs/Local Content"]
    K --> L["Decision Pack Generator\nComparison + Risks"]
    L --> M["Human Approvals\nProcurement/Commercial/PD"]
    M -- "Rejected" --> M1["Revise pack / clarify\nor request new quotes"]
    M1 --> D
    M -- "Approved" --> N["PO Readiness Pack\nSelected supplier + terms"]
    N --> O["Procurement -> Finance Handover Payload"]
    O --> P["Finance Agent\nPO/Invoice/Payment Controls"]

    subgraph Logs ["Audit Trail & Observability"]
        B --> L1["Event Log"]
        F --> L1
        H --> L1
        M --> L1
        O --> L1
    end
```

3.2 Key Interfaces (where devs must implement)

- **Email/Portal Adapter:** intake + send RFQs + track responses
- **Vendor Master / AVL:** supplier metadata + VAT + certifications
- **Document Store:** vendor-scoped RFQ packs
- **Approvals Service:** recorded approvals + approver roles
- **Finance Interface:** PO readiness + payment controls

4) Procurement ↔ Finance data schemas (JSON)

Below are **practical schemas** (not theoretical). Use them as contract payloads between services/agents.

4.1 Procurement → Finance: “PO Readiness Payload”

```
{
```

```

"schema_version": "1.0",
"tender_id": "TND-2026-00091",
"project": {
  "project_id": "KSA-NEOM-ZONEA-001",
  "project_name": "NEOM Zone A Package 1",
  "client_name": "ClientName",
  "currency": "SAR"
},
"selected_supplier": {
  "supplier_id": "SUP-1029",
  "legal_name": "Supplier Legal Name LLC",
  "vat_registration_status": "VERIFIED",
  "vat_number": "300XXXXXXXXX",
  "bank_verification_status": "VERIFIED"
},
"procurement_event": {
  "rfq_id": "RFQ-2026-7781",
  "issue_date": "2026-01-19",
  "quote_received_date": "2026-01-25",
  "quote_valid_until": "2026-02-10"
},
"commercial_terms": {
  "incoterms": "DAP",
  "payment_terms": "30 days from invoice",
  "retention": "5%",
  "warranty": "12 months",
  "delivery_lead_time_days": 21
},
"line_items": [
  {
    "line_id": "1",
    "cost_code": "03-120-450",
    "description": "Rebar supply - Grade X",
    "uom": "ton",
    "quantity": 120,
    "unit_rate": 2450.00,
    "subtotal": 294000.00,
    "vat_applicable": true,
    "vat_rate": 0.15,
    "source_refs": [
      {"type": "boq", "doc": "BOQ.xlsx", "ref": "Sheet1!B22"},
      {"type": "quote", "doc": "SupplierQuote.pdf", "ref": "p3"}
    ]
  }
],
"totals": {
  "subtotal_ex_vat": 294000.00,
  "vat_total": 44100.00,
  "grand_total_inc_vat": 338100.00
},
"approvals": {
  "status": "APPROVED",
  "records": [
    {"role": "Procurement Manager", "approved_by": "user_12",
"approved_at": "2026-01-26T10:12:00Z"},
    {"role": "Commercial Manager", "approved_by": "user_41",
"approved_at": "2026-01-26T11:05:00Z"}
  ]
},
"compliance_report": {
  "rules_passed": ["SA-VAT-01", "SA-SCOPE-01", "SA-PO-01"],

```

```

    "rules_flagged": [],
    "notes": "All compliance checks passed. No scope deviations detected."
  },
  "attachments": [
    {"type": "rfq_pack", "uri": "s3://tenders/TND-2026-00091/vendor/SUP-1029/rfq_pack.zip"},
    {"type": "selected_quote", "uri": "s3://tenders/TND-2026-00091/quotes/SUP-1029.pdf"}
  ],
  "audit": {
    "created_by_agent": "procurement_agent",
    "created_at": "2026-01-26T11:12:00Z",
    "trace_id": "tr_8f2a9c..."
  }
}

```

4.2 Finance → Procurement: “PO Status Update”

```

{
  "schema_version": "1.0",
  "tender_id": "TND-2026-00091",
  "rfq_id": "RFQ-2026-7781",
  "po": {
    "po_number": "PO-4500192231",
    "status": "ISSUED",
    "issued_at": "2026-01-27T08:30:00Z",
    "erp_system": "SAP",
    "erp_ref": "SAP:4500192231"
  },
  "payment_controls": {
    "three_way_match_required": true,
    "invoice_submission_email": "ap@clientdomain.sa"
  },
  "audit": {
    "updated_by": "finance_agent",
    "updated_at": "2026-01-27T08:31:00Z",
    "trace_id": "tr_19b7d2..."
  }
}

```

4.3 Procurement → Finance: “Goods Receipt / Delivery Milestone (Optional)”

```

{
  "schema_version": "1.0",
  "po_number": "PO-4500192231",
  "delivery": {
    "milestone_id": "DEL-01",
    "expected_date": "2026-02-20",
    "actual_date": null,
    "status": "SCHEDULED",
    "site_location": "Riyadh"
  },
  "exceptions": [],
  "audit": {
    "created_by_agent": "procurement_agent",
    "created_at": "2026-01-27T12:10:00Z",
    "trace_id": "tr_22aa01..."
  }
}

```

1 JIRA TICKETS — PROCUREMENT INTELLIGENCE AGENT

(Epics → Stories → Tasks)

EPIC 1: Procurement Agent Core Service

STORY 1.1 – Procurement Agent Intake Gate

Description

Validate approved tender inputs before procurement starts.

Tasks

- Validate Tender ID, project ID, approval status
- Reject unapproved or incomplete tender packages
- Log intake event with trace ID

Acceptance Criteria

- Procurement cannot start without APPROVED status
 - Missing fields trigger a single clarification request
-

STORY 1.2 – Supplier Mapping Engine

Description

Map approved scope packages to eligible suppliers.

Tasks

- Integrate Approved Vendor List (AVL)
- Filter suppliers by trade/category
- Validate supplier status (active/inactive)

Acceptance Criteria

- No RFQ issued to non-approved suppliers
 - Supplier metadata visible (VAT, location, certifications)
-

EPIC 2: RFQ Packaging & Issuance

STORY 2.1 – Vendor-Scoped RFQ Pack Builder

Tasks

- Assemble scope-specific RFQ packs
- Include only relevant drawings/specs
- Redact unrelated information
- Generate RFQ ZIP with manifest

Acceptance Criteria

- Zero cross-vendor document leakage
 - RFQ pack traceable to Tender ID
-

STORY 2.2 – RFQ Issuance & Tracking

Tasks

- Email/portal RFQ issuance
- Due-date tracking
- Automated reminders
- Overdue escalation

Acceptance Criteria

- RFQ lifecycle states: SENT / OPENED / RESPONDED / OVERDUE
 - Reminder frequency configurable
-

EPIC 3: Quote Intake & Normalisation

STORY 3.1 – Supplier Quote Ingestion

Tasks

- Extract pricing, quantities, lead time, exclusions
- Parse PDFs, Excel, Word
- Store original quote read-only

Acceptance Criteria

- 100% traceability to source quote
- Extraction confidence score per field

STORY 3.2 – Quote Normalisation Engine

Tasks

- Align quantities and units
- Map quotes to BOQ lines
- Flag scope gaps and deviations

Acceptance Criteria

- No quote proceeds without normalization
 - Deviations clearly flagged
-

EPIC 4: Saudi Compliance Engine

STORY 4.1 – VAT & ZATCA Compliance Checks

Tasks

- Validate VAT registration
 - Check VAT rate application
 - Flag inconsistencies
-

STORY 4.2 – Localization & Certification Rules

Tasks

- Enforce local content rules (project-based)
 - Validate supplier certifications
 - Track certificate expiry
-

EPIC 5: Approval Orchestration

STORY 5.1 – Approval Workflow Routing

Tasks

- Route to Procurement → Commercial → Project Director
- Track approval delays
- Escalate bottlenecks

EPIC 6: Procurement → Finance Handover

STORY 6.1 – PO Readiness Pack Generator

Tasks

- Build PO payload
 - Validate approval completeness
 - Send structured payload to Finance Agent
-

EPIC 7: Audit & Observability

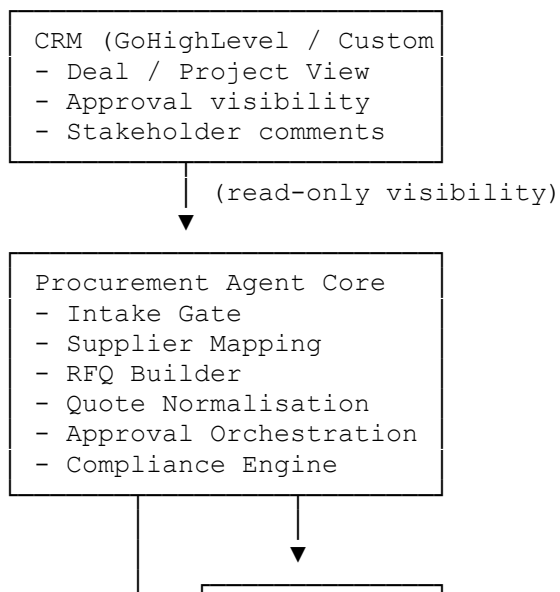
STORY 7.1 – Immutable Audit Logs

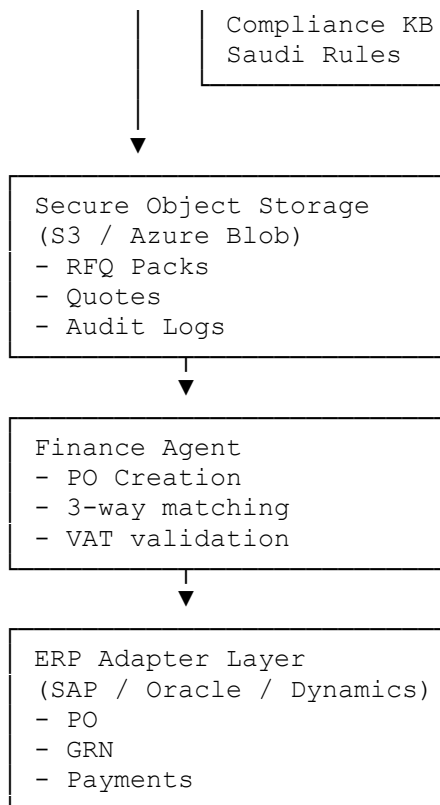
Tasks

- Log RFQs, quotes, approvals, compliance checks
 - Store trace IDs across services
-

2 FULL PROCUREMENT SERVICE ARCHITECTURE

(CRM + ERP + Storage included)





Key Architectural Rules

- CRM = **visibility & approvals**, not pricing logic
- Procurement Agent = **intelligence + control**
- ERP = **execution system only**
- Storage = **single source of truth**

3 \$ SAUDI PROCUREMENT COMPLIANCE RULE ENGINE

(Deterministic pseudo-code)

```
function run_procurement_compliance(quote, supplier, project):  
  
    flags = []  
  
    // VAT / ZATCA  
    if supplier.vat_required and supplier.vat_number is null:  
        flags.add(HIGH, "SA-VAT-01", "Supplier VAT not verified")  
  
    if quote.vat_rate != 0.15 and quote.vat_applicable:  
        flags.add(MEDIUM, "SA-VAT-02", "Incorrect VAT rate")  
  
    // Localization  
    if project.requires_local_content:
```

```
        if supplier.local_content_pct < project.min_local_content:
            flags.add(HIGH, "SA-LOC-01", "Local content requirement not met")

// Certifications
for cert in project.required_certifications:
    if cert not in supplier.certificates or cert.expired:
        flags.add(HIGH, "SA-CERT-01", "Missing or expired certification")

// Scope integrity
for boq_line in project.boq:
    if boq_line not in quote.items:
        flags.add(HIGH, "SA-SCOPE-01", "BOQ line missing in quote")

// Payment terms
if quote.payment_terms != project.allowed_payment_terms:
    flags.add(MEDIUM, "SA-COMM-01", "Payment terms deviation")

// Decision
if any(flags.severity == HIGH):
    status = "BLOCKED"
else:
    status = "REVIEW"

return {status, flags}
```
